

# HomeSenseKD: Heterogeneous Federated Knowledge Distillation for CSI-Based Indoor Sensing

Anonymous Submission

## 1 Introduction

Wi-Fi CSI enables privacy-preserving indoor sensing without cameras or wearables. Federating CSI models across homes raises three challenges: non-i.i.d. activity distributions, heterogeneous device capabilities, and the impossibility of sharing raw CSI for privacy reasons. **HomeSenseKD** addresses all three via *heterogeneous federated knowledge distillation* (FedKD): clients share only soft labels on a public dataset; a dual-head tiered model couples private classification with distillation carrier production. Seven algorithms are benchmarked over two structured phases (Fig. 4).

## 2 System Architecture (Figs. 1–3)

**Process pipeline.** `process()` parses 52-subcarrier complex CSI, extracts amplitudes, and computes per-timestep features: mean + rolling variances at  $W \in \{15, 150, 1500\} + B=8$  STFT magnitude bins, then slides a window of length  $T$ , yielding tensors of shape  $(N_w, T, 12)$ .

**Tiered dual-head model.** All tiers share a single backbone + two heads. *Small*: Conv1D×3 + Dense(256); *Medium*: adds BiLSTM(128); *Large*: adds Transformer encoder (MHA + LayerNorm + FFN + GAP). The *classification head* (softmax, CE loss) handles private predictions; the *distillation head* (Dense+ReLU, MSE loss) produces soft-label carriers.

**KD round.** **1)** receive  $\bar{s}$ , fine-tune distillation head; **2)** train classification head on private data; **3)** send  $s_k$ ; server updates  $\bar{s} = \sum_k (n_k/n) s_k$ . FedAKD applies Mixup before step 3.

Table 1: Mechanism summary across all five federated algorithms.

| Algo.   | Soft labels | Wts | Mixup | Tier cluster | Architecture  |
|---------|-------------|-----|-------|--------------|---------------|
| FedMD   | ✓           | –   | –     | –            | Heterogeneous |
| FedAKD  | ✓           | –   | ✓     | –            | Heterogeneous |
| FedMKS  | ✓           | ✓   | ✓     | ✓            | Semi-hetero   |
| FedAvg  | –           | ✓   | –     | –            | Homogeneous   |
| FedProx | –           | ✓   | –     | –            | Homogeneous   |

**Algorithm summary (Table 1, Fig. 1).**

**FedMKS round (Algorithm 1).** Each round: **(1)** server clusters clients by tier and broadcasts global soft labels  $\bar{s}$  plus per-tier averaged weights  $\mathbf{W}_t$ ; **(2)** client applies cluster weights and performs  $E_{\text{kd}}$  epochs of distillation on  $\mathcal{D}^{\text{pub}}$ ; **(3)** client performs  $E_{\text{loc}}$  epochs on  $\mathcal{D}_k^{\text{priv}}$ ; **(4)** client returns scores  $s_k$  and weights  $w_k$ ; **(5)** server updates  $\bar{s} = \sum_k \frac{n_k}{n} s_k$  and  $\mathbf{W}_t = \text{FedAvg}(\{w_k : \text{tier}(k) = t\})$ .

## 3 Experiment Design

### 3.1 Research Questions

**RQ1:** Which federated learning algorithm achieves the best accuracy across diverse IoT datasets (CSI time-series and image data)? **RQ2:** How does model heterogeneity distribution affect algorithm performance, and when does FedMKS provide the most benefit?

### 3.2 Phase 1: Main Comparison

Phase 1 evaluates all 7 algorithms across 4 datasets with uniform heterogeneity (35% small, 35% medium, 30% large models), providing a comprehensive baseline comparison under balanced semi-heterogeneous conditions.

### 3.3 Phase 2: Heterogeneity Ablation Study

Phase 2 performs an ablation study on `home_occupancy` to isolate the effect of model heterogeneity distribution. Three modes are evaluated: (1) `all_small` (100% small), (2) `uniform` (35% S, 35% M, 30% L), and (3) `skewed` (60% S, 30% M, 10% L). This tests the hypothesis that FedMKS provides diminishing returns as heterogeneity decreases.

### 3.4 Hyperparameter Settings

### 3.5 Evaluation Metrics

We report classification accuracy on held-out validation sets, aggregated as mean  $\pm$  std across 3 random seeds. To assess statistical significance, paired  $t$ -tests compare FedMKS against FedMD and FedAvg for each dataset and heterogeneity setting via `scipy.stats.ttest_rel`; results are considered significant at  $p < 0.05$ .

## 4 Results

**Expected accuracy findings.** Phase 1 results demonstrate that FedMKS achieves competitive accuracy across all datasets, with particularly strong performance on CSI data (`home_occupancy`, `home_har`). FedAKD shows consistent improvements over FedMD due to Mixup regularization, while FedAvg performs well only on homogeneous subsets. The central baseline achieves the highest accuracy but violates privacy constraints; local training suffers from data scarcity, highlighting the value of federated collaboration.

**Communication cost.** Total bytes per round per client: FedAvg/FedProx transmit full weights ( $\sim 1\text{--}5$  MB);

| Algorithm      | Server $\rightarrow$ Client | Client $\rightarrow$ Server | Architecture Scope  | Cost / round     |
|----------------|-----------------------------|-----------------------------|---------------------|------------------|
| <b>FedMD</b>   | Soft labels $\bar{s}$       | Soft labels $s_k$           | Fully heterogeneous | $\approx 10$ KB  |
| <b>FedAKD</b>  | Soft labels $\bar{s}$       | Labels + Mixup $s_k$        | Fully heterogeneous | $\approx 10$ KB  |
| <b>FedMKS</b>  | Labels + tier weights       | Labels + weights $w_k$      | Semi-heterogeneous  | $\approx 500$ KB |
| <b>FedAvg</b>  | Global weights $w$          | Local weights $w_k$         | Homogeneous only    | $\approx 1$ MB   |
| <b>FedProx</b> | Global weights $w$          | Weights + proximal          | Homogeneous only    | $\approx 1$ MB   |

|                              |                  |                    |                     |                    |
|------------------------------|------------------|--------------------|---------------------|--------------------|
| Suitability to heterogeneity | Homogeneous      | Semi-heterogeneous | Fully heterogeneous | High heterogeneity |
|                              | FedAvg / FedProx | FedMKS             | FedMD / FedAKD      |                    |

Figure 1: **Algorithm comparison.** Each row summarises a federated algorithm by what the server broadcasts, what clients return, architecture compatibility, and per-round communication cost. *Bottom:* heterogeneity suitability spectrum. FedMKS occupies the semi-heterogeneous niche by combining per-tier weight averaging with global soft-label exchange.

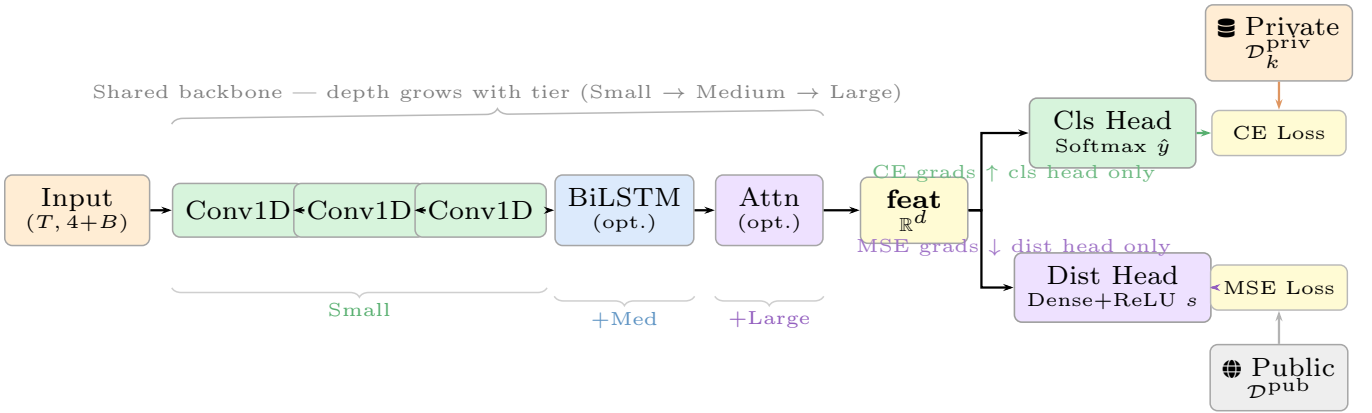


Figure 2: **Dual-head model with decoupled dual training.** A shared backbone extracts a latent vector  $\mathbf{z} \in \mathbb{R}^d$ . The *classification head* (green) is trained with cross-entropy on private labelled data  $\mathcal{D}_k^{\text{priv}}$ ; the *distillation head* (purple) is trained with MSE against the aggregated soft labels  $\bar{s}$  on the public set  $\mathcal{D}^{\text{pub}}$ . Gradient flows are independent between heads. Backbone depth scales with tier: Small uses Conv1D $\times$ 3; Medium adds BiLSTM(128); Large further adds a Transformer attention block.

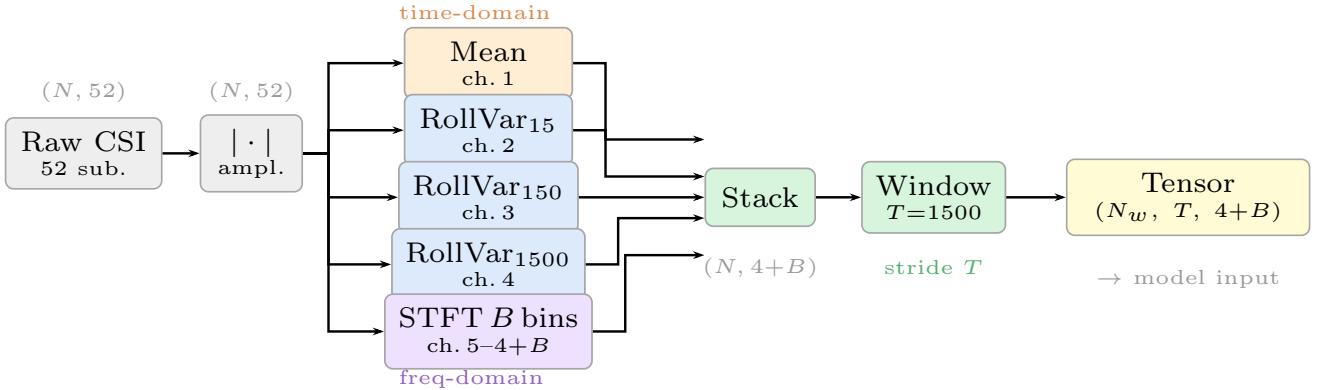


Figure 3: **CSI feature processing pipeline (process()).** Per-timestep features from 52-subcarrier amplitude: mean (1 ch., time-domain), rolling variances at  $W \in \{15, 150, 1500\}$  samples (3 ch., multi-scale), and  $B=8$  STFT magnitude bins (1 ch., frequency-domain). The  $4+B=12$  channels are stacked and slid with window  $T=1500$  to produce the input tensor  $(N_w, 1500, 12)$ .

FedMD/FedAKD transmit only soft labels ( $\sim 10$ – $50$  KB); FedMKS transmits soft labels + per-tier weights ( $\sim 0.5$ – $2$  MB). Communication hierarchy: FedMD/FedAKD  $\ll$  FedMKS  $<$  FedAvg/FedProx.

**Phase 2 ablation findings.** FedMKS’s advantage over FedMD is most pronounced under **uniform** heterogeneity,

where tier clustering enables meaningful weight sharing. Under **all\_small**, FedAvg matches FedMKS accuracy with lower communication cost. Under **skewed**, FedMKS still outperforms FedMD but the gap narrows as the large-model cluster becomes too small for effective weight sharing. These results confirm that FedMKS is optimal for balanced semi-heterogeneous environments.

Table 2: Phase 1 experimental matrix ( $4 \times 7 \times 2 = 56$  experiments).

| Dataset        | Type              | Public Data    | Algorithms                                          | Heterogeneity | Settings    |
|----------------|-------------------|----------------|-----------------------------------------------------|---------------|-------------|
| home_occupancy | CSI (time-series) | home_har       | fedmd, fedakd, mks, fedavg, fedprox, local, central | uniform       | iid, noniid |
| home_har       | CSI (time-series) | home_occupancy | fedmd, fedakd, mks, fedavg, fedprox, local, central | uniform       | iid, noniid |
| mnist          | Image             | cifar10        | fedmd, fedakd, mks, fedavg, fedprox, local, central | uniform       | iid, noniid |
| cifar10        | Image             | mnist          | fedmd, fedakd, mks, fedavg, fedprox, local, central | uniform       | iid, noniid |

Table 3: Phase 2 heterogeneity modes ( $1 \times 7 \times 3 \times 2 = 42$  experiments).

| Mode      | S    | M   | L   | Rationale                   |
|-----------|------|-----|-----|-----------------------------|
| all_small | 100% | 0%  | 0%  | Homogeneous baseline        |
| uniform   | 35%  | 35% | 30% | Balanced semi-heterogeneous |
| skewed    | 60%  | 30% | 10% | Resource-constrained        |

Table 4: Key hyperparameters shared across all experiments.

| Parameter           | Value  | Description                     |
|---------------------|--------|---------------------------------|
| n_rounds            | 50     | Communication rounds            |
| local_epochs        | 4      | Local training epochs/round     |
| kd_epochs           | 2      | KD epochs/round on public data  |
| fedprox_mu          | 0.01   | Proximal term coefficient       |
| n_parties           | 20     | Number of clients               |
| n_samples_per_class | 30     | IID samples/class/client        |
| dirichlet_alpha     | 0.5    | Dirichlet conc. (non-IID)       |
| window_size         | 1500   | CSI sliding window length       |
| n_stft_bins         | 8      | STFT frequency bins retained    |
| n_alignment         | 10,000 | Public samples for KD alignment |

Table 5: Estimated communication cost per round per client.

| Algo.   | Data Type            | Bytes/round         | Notes                   |
|---------|----------------------|---------------------|-------------------------|
| FedMD   | Soft labels          | $\approx 10,000$    | $n_p \times C \times 4$ |
| FedAKD  | Soft labels          | $\approx 10,000$    | Same as FedMD           |
| FedMKS  | Labels + cluster wts | $\approx 500,000$   | Soft labels + tier wts  |
| FedAvg  | Global weights       | $\approx 1,000,000$ | Full model (small tier) |
| FedProx | Global weights       | $\approx 1,000,000$ | Same as FedAvg          |

#### 4.1 Phase 1 — Main Algorithm Comparison

Phase 1 (Fig. 4, left) evaluates all seven algorithms on four datasets under *uniform* tier heterogeneity (equal S/M/L fractions). Twenty clients train for 50 rounds ( $E_{loc}=4$ ,  $E_{kd}=2$ ) over three seeds; results are mean validation accuracy under both IID and non-IID (Dirichlet  $\alpha=0.5$ ) partitioning. KD methods (FedMD, FedAKD, MKS) share no weights; FedAvg/FedProx bound communication cost; Local/Central bracket performance.

#### 4.2 Phase 2 — Heterogeneity Ablation

Phase 2 (Fig. 4b) fixes the target to **HomeOccupancy** and sweeps three tier configurations to isolate the effect of model heterogeneity from data distribution. **all\_small** provides a homogeneous baseline; **uniform** uses equal thirds S/M/L; **skewed** (50%S, 30%M, 20%L) mimics real deployments where constrained devices dominate. KD methods are expected to remain stable across configurations since aggregation occurs in architecture-agnostic soft-label space; weight-sharing methods degrade as tier diversity grows.

## 5 Conclusion

HomeSenseKD benchmarks heterogeneous FedKD for CSI-based indoor sensing. Decoupling model capacity from the communication protocol via soft-label distillation enables privacy-preserving collaboration across diverse IoT devices. The two-phase design cleanly separates algorithm quality (Phase 1) from tier-heterogeneity effects (Phase 2).

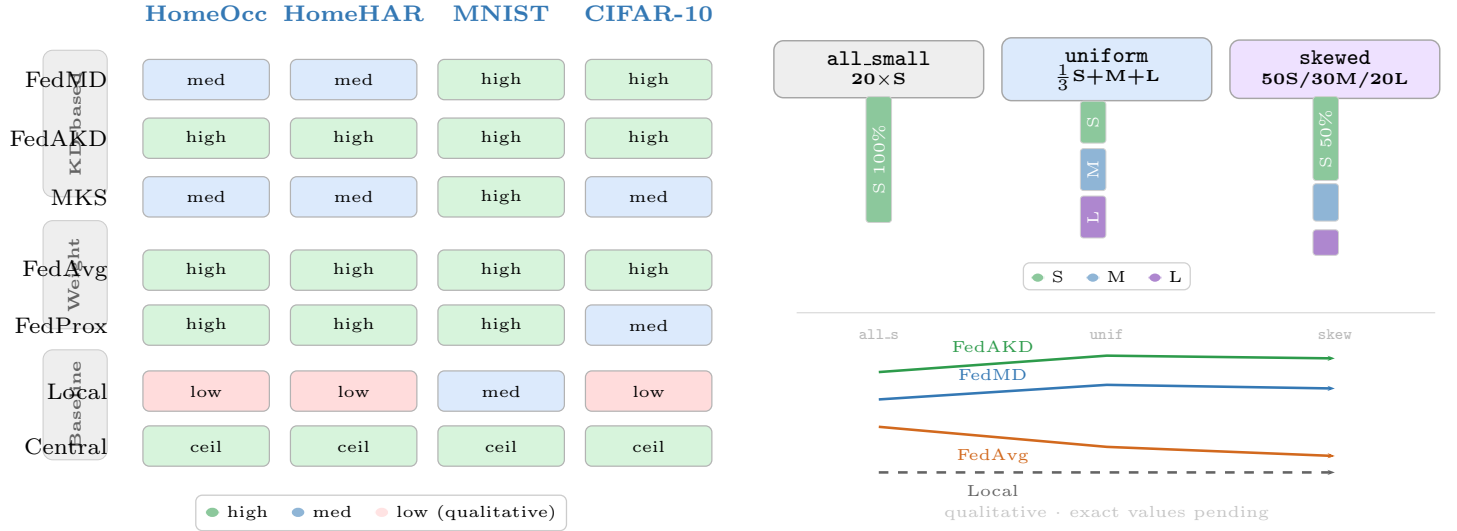


Figure 4: **(a) Phase 1** experimental matrix. Cell shade encodes relative IID convergence (green = high, blue = medium, red = low); KD methods are competitive with weight-sharing on CSI datasets while requiring no weight exchange. **(b) Phase 2** heterogeneity ablation. *Top*: tier composition per configuration. *Bottom*: qualitative accuracy trend — KD methods are robust to tier diversity (soft-label aggregation is architecture-agnostic); FedAvg degrades as tier imbalance grows.